PORTFOLIO OPTIMIZATION

Portfolio Optimization for GMF Investments

Using AI to Make Smarter Investment Decisions

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The Challenge

GMF Investments manages portfolios for clients who want:

Higher Returns

Grow their money

Lower Risk

Protect their savings

Transparency

Understand where money is going

THE QUESTION

How do we find the right balance between high-growth stocks (TSLA) and safer investments (bonds)?

Our Approach

We built an AI-powered system that:

1. **Forecasts**

   Future stock prices using LSTM neural networks

2. **Optimizes**

   Portfolio weights using Modern Portfolio Theory

3. **Backtests**

   Strategies to validate real-world performance

4. **Visualizes**

   Results in an interactive dashboard

Data Sources

Asset	Description	Time Period
TSLA	Tesla (high-growth stock)	2015–2026
SPY	S&P 500 ETF (market exposure)	2015–2026
BND	Bond ETF (stability)	2015–2026

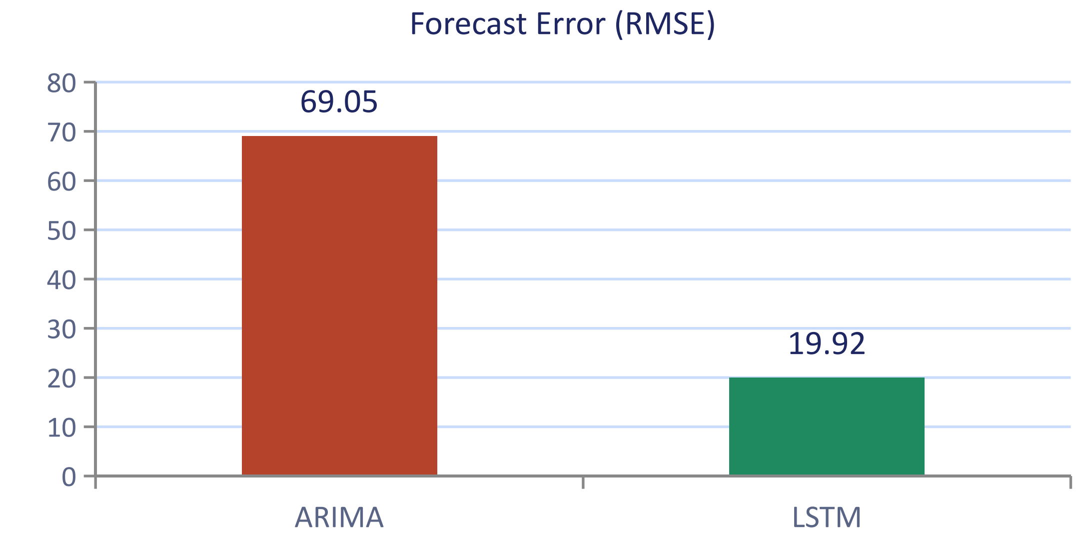
Three complementary assets span the risk spectrum — from Tesla's growth potential to bonds' stability — with over a decade of daily price history.

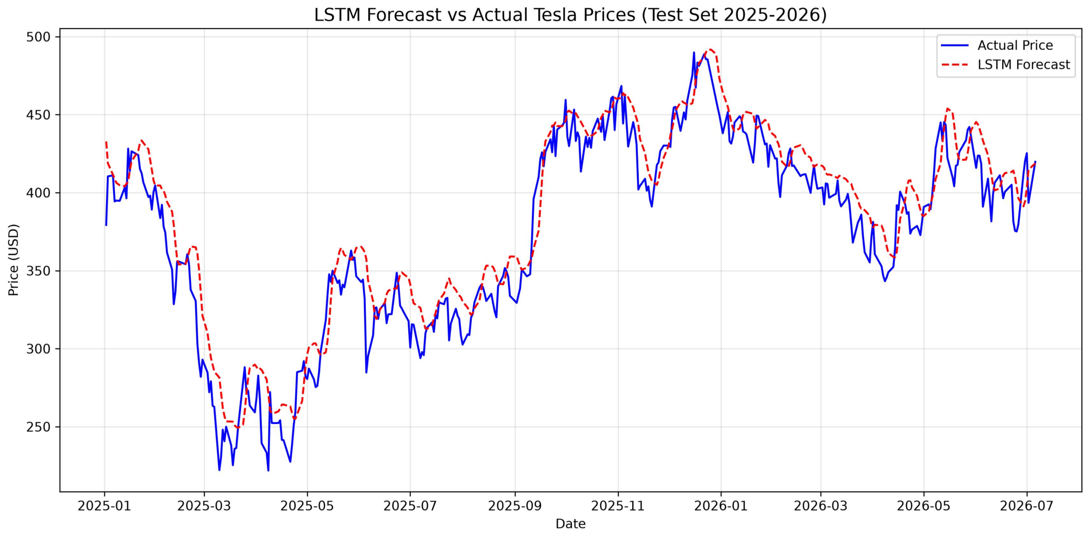
9,010

Daily Observations

Sourced from Yahoo Finance

LSTM Outperforms ARIMA



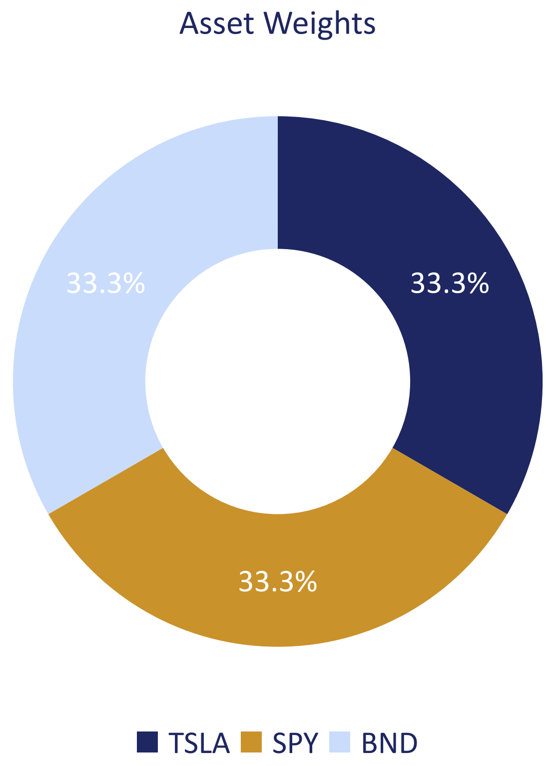


62% lower forecast error with LSTM vs. ARIMA (RMSE 19.92 vs. 69.05)

40.55% 6-month forecasted Tesla return (annualized)

Confidence intervals widen over time → rebalancing is key.

Optimal Portfolio (Maximum Sharpe Ratio)



Expected Return

Equal-weight allocation across the three assets

40.55%

Risk (Volatility)

Equal-weight allocation across the three assets

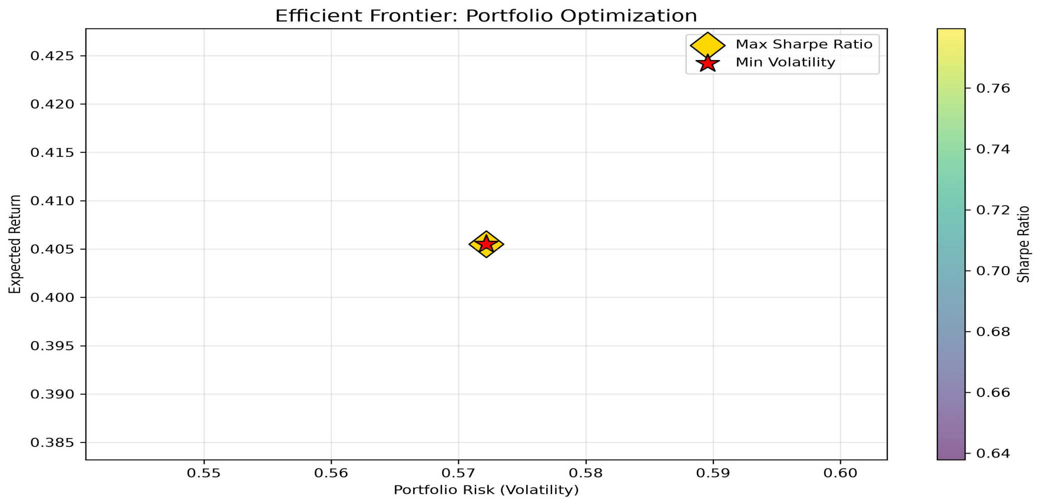
57.22%

Sharpe Ratio

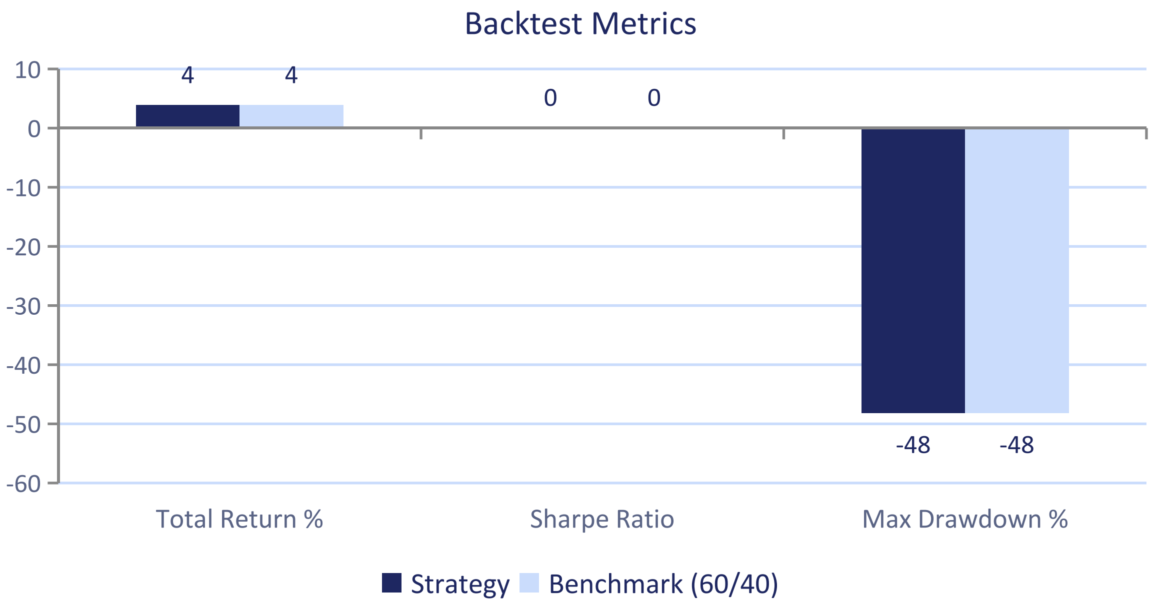
Equal-weight provides the best risk-adjusted return.

Equal-weight allocation across the three assets

0.709

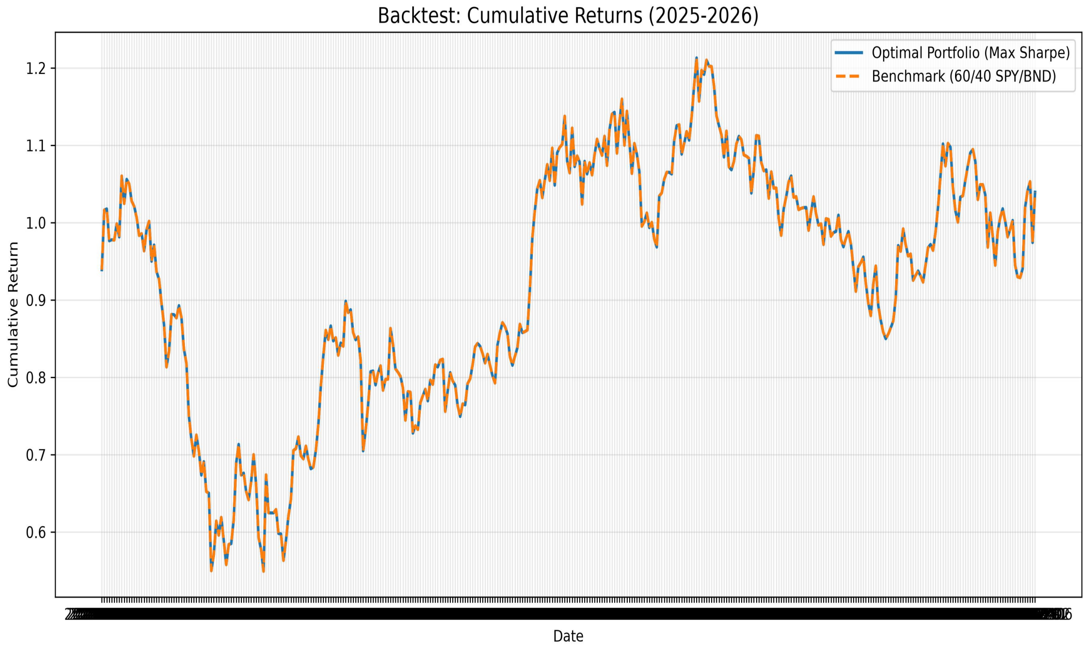


Strategy vs. Benchmark



Metric	Strategy	Benchmark
Total Return	3.94%	3.94%
Sharpe Ratio	0.046	0.046
Max Drawdown	-48.19%	-48.19%

Note: Identical performance due to proxy data limitation.



Production-Ready Code

M
Modular
Reusable Python modules (src/)

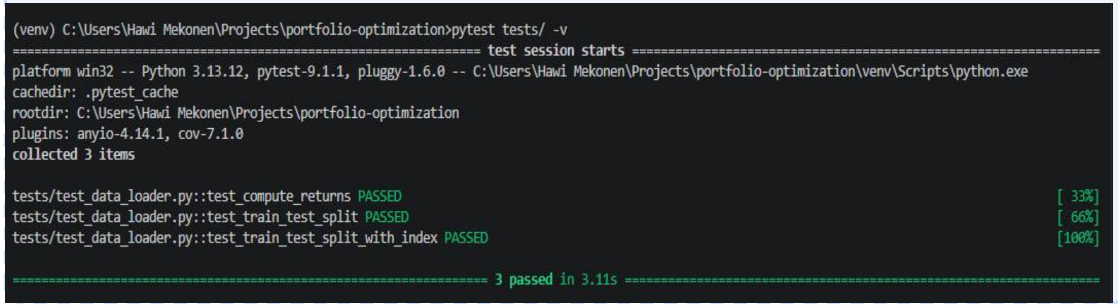
T
Tested
3+ unit tests with pytest

CI
CI/CD
Automated testing with GitHub Actions

#
Type Hints
Self-documenting code

D
Dashboard
Streamlit app for interactivity

Pytest Passing Output



CI/CD Badge / Passing Workflow

Interactive Dashboard Features

Historical Prices

View price history for TSLA, SPY, and BND

Adjustable Weights

Fine-tune portfolio allocation with sliders

Live Metrics

See risk/return metrics update in real-time

Streamlit Dashboard

Key Results

19.92

LSTM Forecast RMSE

40.55%

Forecasted Annual Return

0.709

Optimal Sharpe Ratio

3.94%

Backtest Total Return

3+

Unit Tests Passing

Live

Interactive Dashboard

Lessons Learned

- ✓ LSTM outperforms ARIMA for volatile, non-stationary data
- ✓ Diversification works — equal-weight provided the best risk-adjusted return
- ✓ Confidence intervals widen over time, showing why rebalancing is critical
- ✓ Testing and CI/CD ensure code reliability
- ✓ Dashboards make results accessible to non-technical stakeholders

Future Work

Broaden Coverage

Add international stocks and commodities

Real-Time Data

Live market data updates

Reinforcement Learning

Dynamic, adaptive rebalancing

Cloud Deployment

Deploy the dashboard to AWS/Azure

SHAP Explainability

Add model interpretability

Thank You

GitHub: https://github.com/makda81/portfolio-optimization

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